

netic energy. These parameters may cause unwanted effects such as electromagnetic interference (EMI) and could physically damage the operational equipment.

Anechoic RF chambers are set up for electromagnetic compliance (EMC) testing, and the furniture inside the room is made of non-metals. "It is a completely different environment covered with observers," says Varma. Observers are placed in the room to monitor test data. These provide us with frequency versus amplitude analysis, and the data can be checked for compliance.

Testing wireless parameters. Going wireless not only results in fewer wires and easy usability, it also brings the issue of radiation exposure. So the devices have to be tested for radiation, too. The most common use of wireless technology in our phones is for making phone calls. Testing for proper GSM standard has to be performed in every device. "We use EMF receivers for testing electromagnetic parameters. These provide a graphical representation for frequency versus amplitude analysis," says Varma. Anritsu Me7834L and R&S CMW500 are commonly used for testing GSM signals.

SAR is another area of concern with mobile devices. 1.69W/kg is the maximum radiation allowed under many compliance standards. The most accepted method for SAR testing involves simulating a human. The simulated body consists of an orthogonal probe mounted on a robot arm to measure RF fields transmitted into a calibrated tissue stimulant, which is a fluid used to simulate brain or muscle tissue.

SOME RECENT EQUIPMENT	
Equipment	Test Category
Anritsu E series	Remote radio heads installed atop 4G towers
Keysight M937xA PXle Multiport VNA	High-speed transfer bus
ART-Fi and R&S Compliance Bot with ART-MAN	SAR measurement
Omega OM-DAQXL	Data logger
Aspen Electronics Sefram DAS 50	Data logger
Testo Saveris 2	Thermal data logger

Quality of the fluid used to simulate the brain or muscle tissue is obviously important, as its electrical properties vary with frequency, temperature and age. With this method, wireless devices are tested at the highest certified power level in laboratory conditions by a SAR test system. Some common SAR test equipment are ART-MAN from Art-Fi and cSAR3D from SPEAG.

A lot of data to be logged. Another equipment commonly used in testing is a data logger. With capabilities including automated data collection, it can be activated and left unattended for a specified period prior to deployment. Electronic data loggers have replaced chart recorders in many applications. Data loggers are becoming increasingly common for specific purposes that require monitoring for longer durations.

Temperature measurement, for example, requires devices to be operational for long hours. Data loggers have been a life-saver here. Simply connect a thermometer to the data logger and you are good to go, literally!

SDI-12 protocol allows some instruments to be connected to a variety of data loggers. MODBUS and Canbus are some other protocols being used in today's data

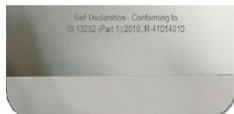


Fig. 2: IS conformity is displayed on back covers now

loggers. Some data loggers use a flexible scripting environment to adapt themselves to various non-standard protocols.

What to look ahead to

IS 13252 is focused on the safety of IT and electronic equipment, whereas other standards like IS 616 focus on the safety requirements for mains-operated electronic and related apparatus for household and similar general use. However, test equipment used here are similar throughout the spectrum of standards. The type of equipment, however, is moving towards automation. Data loggers provide options like deploy-and-forget, where you have the option of going through measured data at your own pace.

With connectivity being the focus in equipment development, the original model of a stand-alone data logger is changing to one of a device that collects data but also has access to wireless communications. We are moving away from proprietary products with commercial software to open source software and hardware devices.

In future, specialised high-speed digital data loggers could replace traditional instruments such as oscilloscopes and logic analysers. But we are still miles away from it! **RFY**

Products covered under IS 13252

Laptops, notebooks, tablets, visual display units, video monitors of screen size 81.2cm (32-inch) and above, printers, plotters, scanners, wireless keyboards, telephone answering machines, set-top boxes, automatic data processing machines, power adaptors for IT equipment, mobile phones, cash registers, point-of-sale terminals, copying machines/duplicators, smartcard readers, mail processing machines/postage machines/franking machines, passport readers and power banks for use in portable applications, among others, are the products covered under IS 13252.